IS1204 Fundamentals of Software Engineering

H N D Thilini

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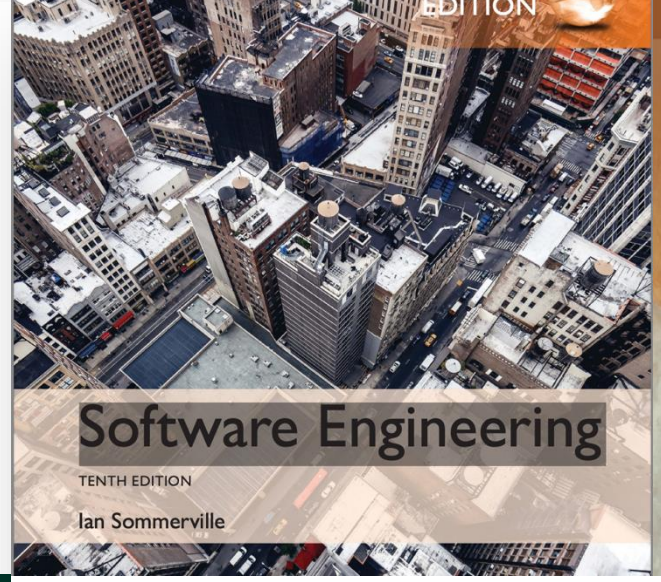
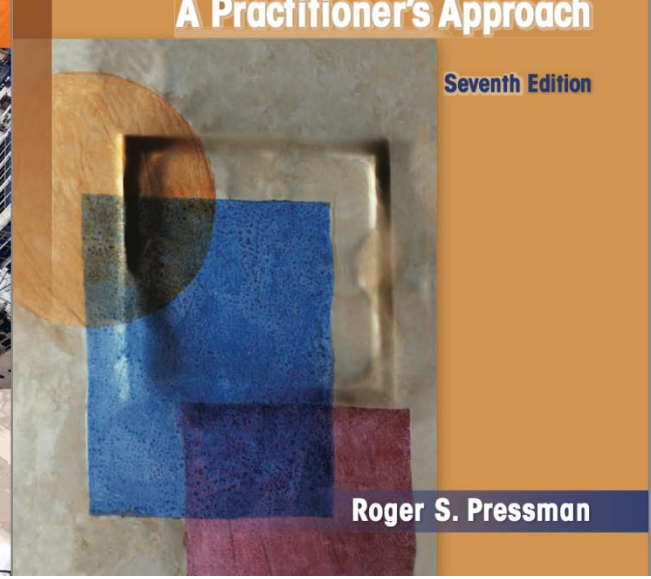
Recommended Reading

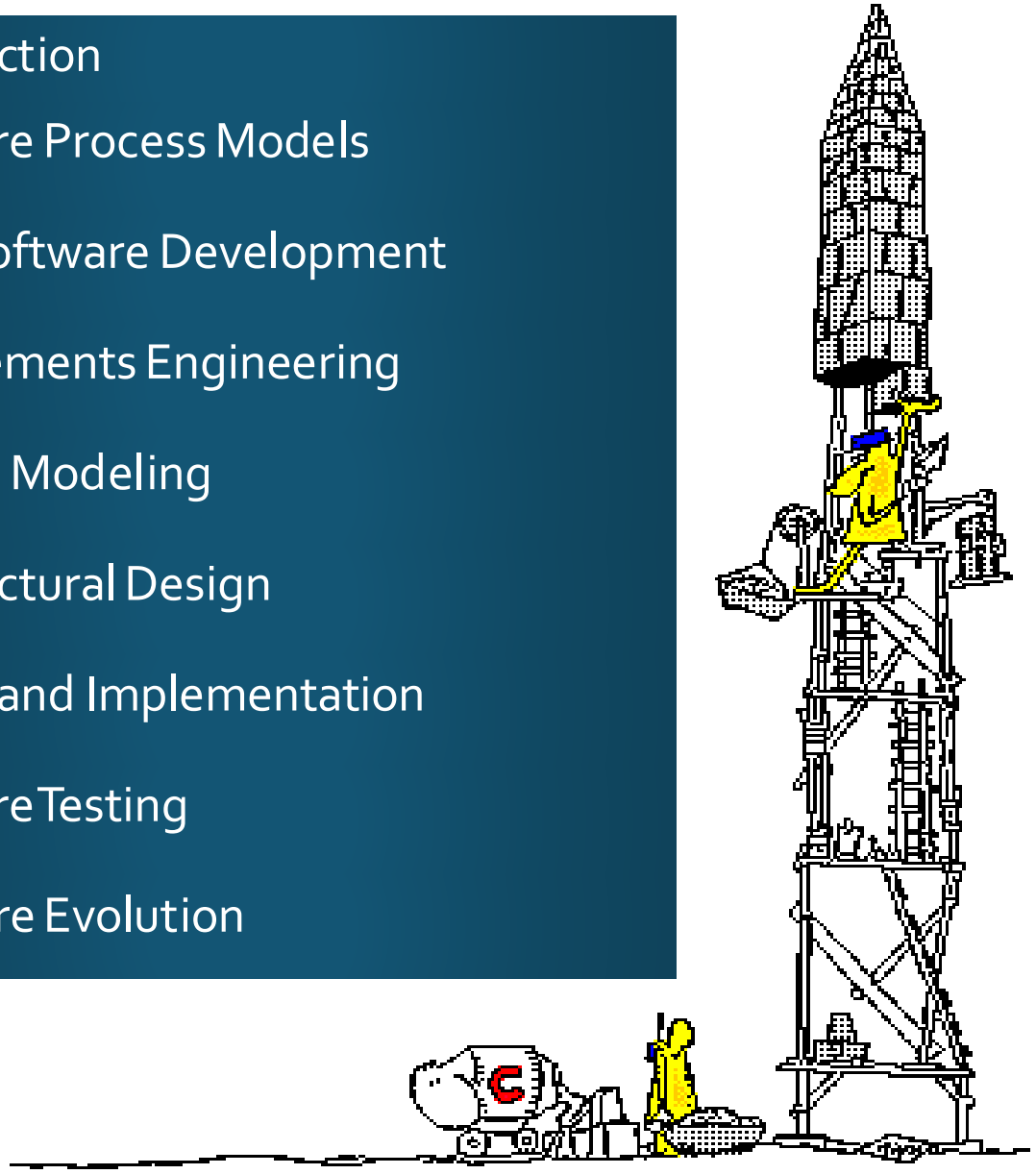




- Ian Sommerville, Software Engineering, 10th Edition, Pearson, 2017.
- R Pressman, Software Engineering – A Practitioners Approach, 7th Edition, McGraw Hill.

Course Content

Chapter 1	Introduction
Chapter 2	Software Process Models
Chapter 3	Agile Software Development
Chapter 4	Requirements Engineering
Chapter 5	System Modeling
Chapter 6	Architectural Design
Chapter 7	Design and Implementation
Chapter 8	Software Testing
Chapter 9	Software Evolution



Course Structure

2 Credit Course

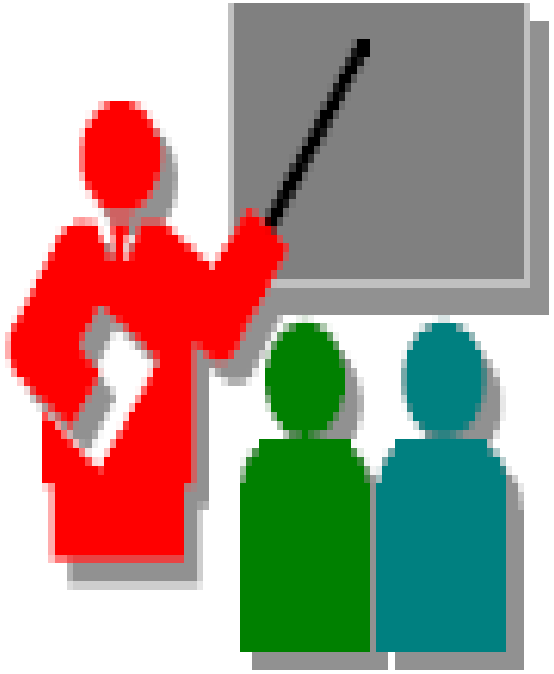
- Lectures (Friday – 10.00 am to 12.00 noon)

Course Assessment:

- Exam paper: 2 hours (70%)
- Assignments: In-class + Take home (30%)

Lecture Notes / Reading Materials / Notices:

- Published in UG-VLE



Overall Objective

After successfully completing this subject, students should be able to:

- Understand what is software engineering, associated principles and concepts.
- Evaluate different software processes / techniques in systematic development of quality software products.
- Apply software engineering principles and techniques appropriately to design moderately complex software system.

1. Introduction

Learning Outcomes:

- Define what software engineering is and why it is important
- Identify suitable software engineering techniques for development of different types of software
- Recognize the ethical and professional issues that are important for software engineers

What we are going to discuss

1.1 Software

- What is software?
- Types of software
- Characteristics of software

1.2 Software Engineering

- What is software engineering?
- Systems Engineering & Software Engineering
- Software engineering costs
- What are the key challenges facing software engineering?
- Professional Practice

What is Software?

Many people think that software is simply another word for **computer programs**.

Computer Program?

- Set of instructions
- Enables a computer to perform a specific task

What is Software?

Many people think that software is simply another word for **computer programs**.

Software ≠

Computer Program

What is Software?

- Software is a combination of
  - Computer programs
  - System Documentation (documents produced during development)
  - User Documentation (delivered with programs to customer at the time of release).

Computer Program

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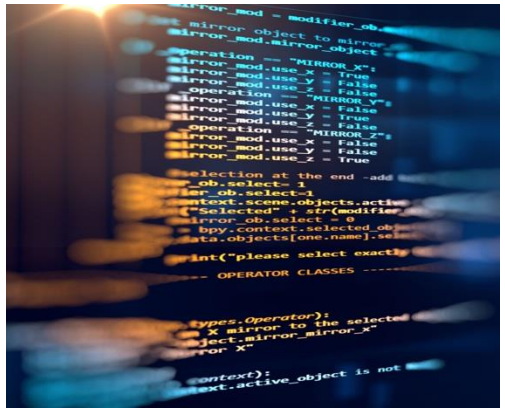
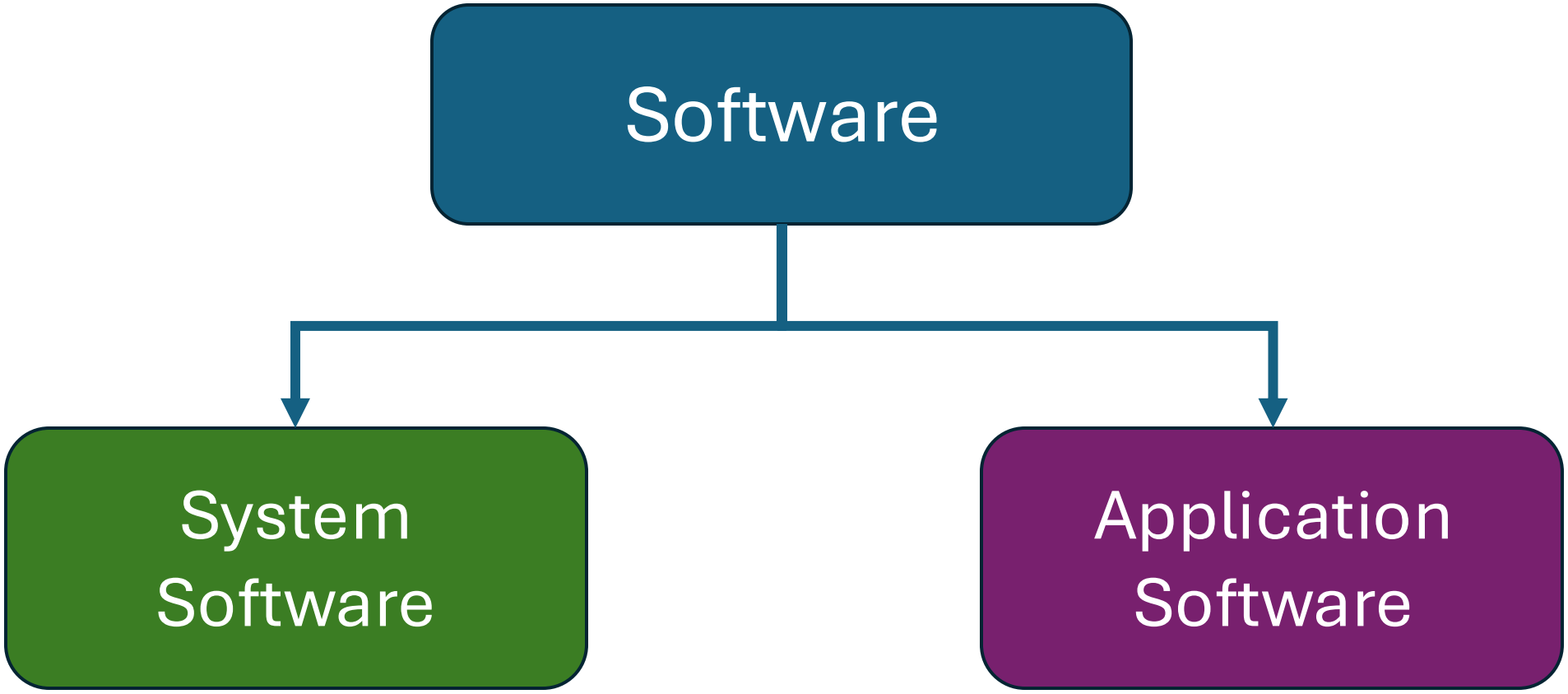
System and User Documentations



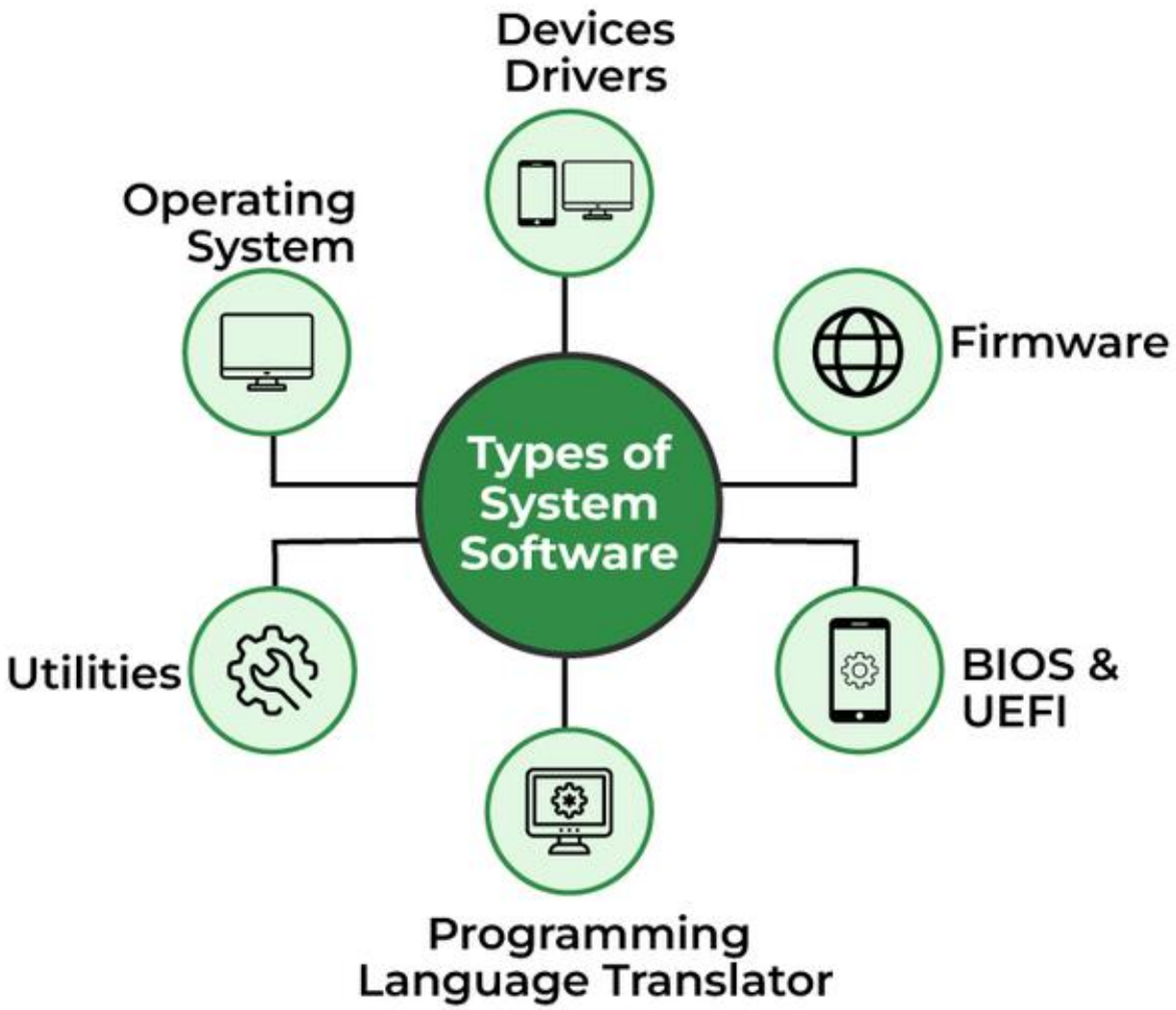
Software Types



System Software

- low-level software that manages and controls a computer's hardware and provides basic services to higher-level software
- provides a platform for other software
- consists of programs written in low-level languages, used to interact with the hardware at a very basic level
- serves as the interface between the hardware and the end users

Types of System Software

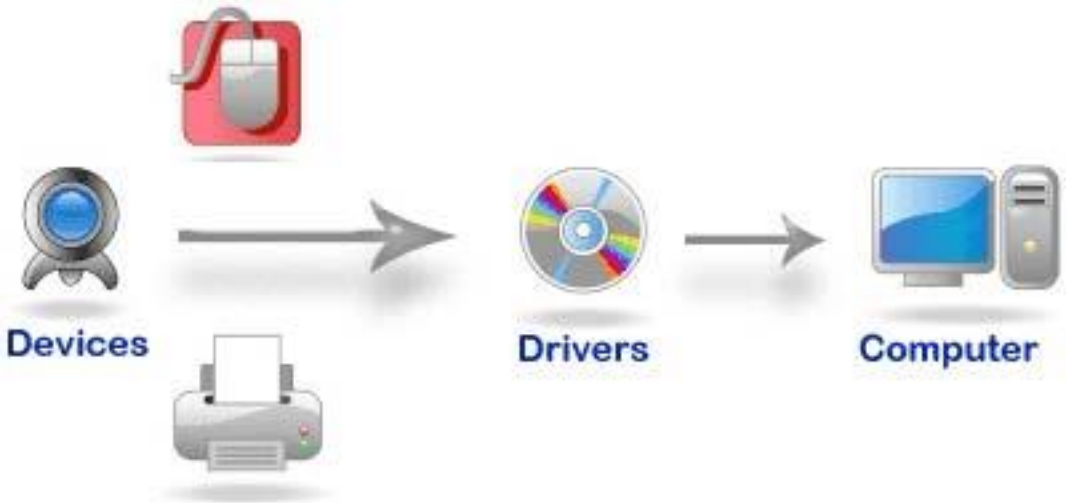


System Software Types

- Operating Systems
  - each operating system is different - most provide a Graphical User Interface (GUI) through which user can manage files and folders and perform other tasks
  - a collection of software that handles resources and provides general services for the other applications that run on a specific system
  - Every device, whether a desktop, laptop, or mobile phone, requires an operating system to provide its basic functionality

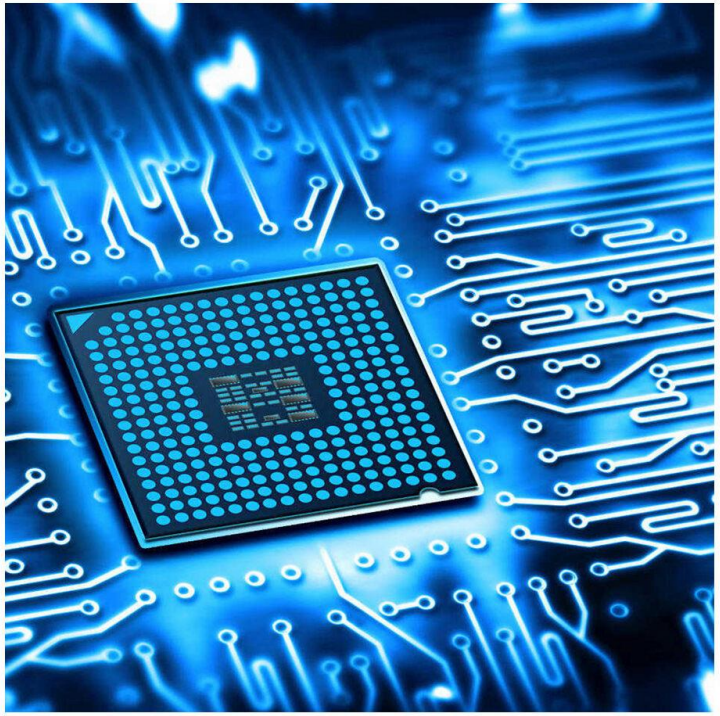
System Software Types

- Device Drivers
  - controls particular hardware attached to the system
  - Hardware devices that need a driver to connect to a system include displays, sound cards, printers, mouse, and hard disks
  - Examples:
    - BIOS Driver
    - Display Drivers
    - Motherboard Drivers
    - Printer Drivers
    - ROM Drivers
    - Sound card Driver
    - USB Drivers
    - USB Drivers
    - VGA Drivers
    - VGA Drivers
    - Virtual Device Drivers
    - …



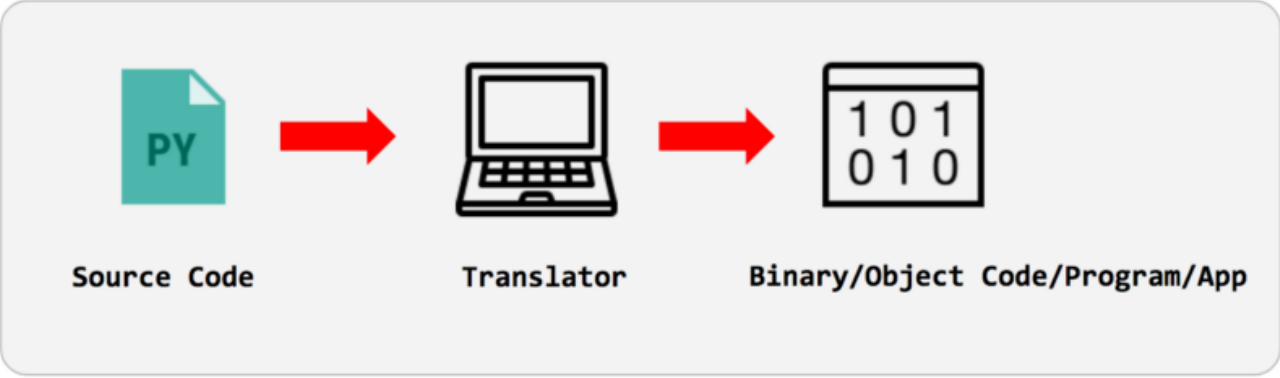
System Software Types

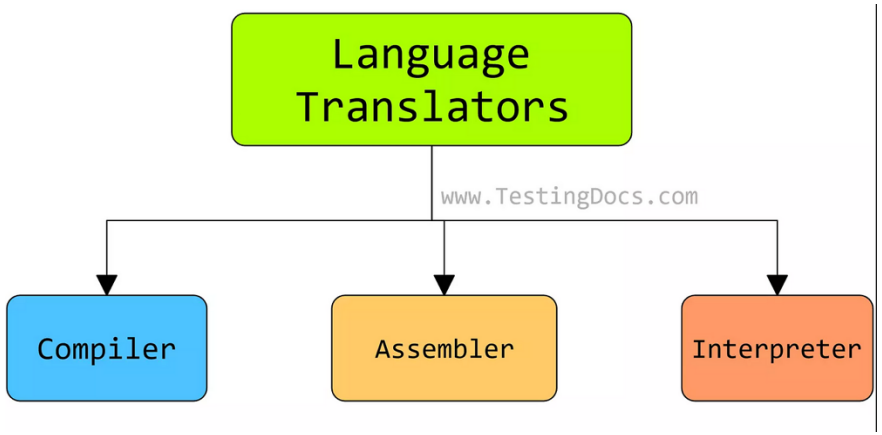
- Firmware
  - permanent software embedded into a read-only memory.
  - set of instructions permanently stored on a hardware device.
  - provides essential information regarding how the device interacts with other hardware.
  - can be considered as ‘semi-permanent’ as it remains permanent unless it is updated using a firmware updater.
  - Examples:
    - BIOS
    - Computer Peripherals
    - Consumer Applications
    - Embedded Systems
    - UEFI



System Software Types

- Programming Language Translators
  - mediator programs on which software programs rely to translate high-level language code to simpler machine-level code.
  - Besides simplifying the code, the translators also do the following :
    - Assign data storage
    - Enlist source code as well as program details
    - Offer diagnostic reports
    - Rectify system errors during the runtime

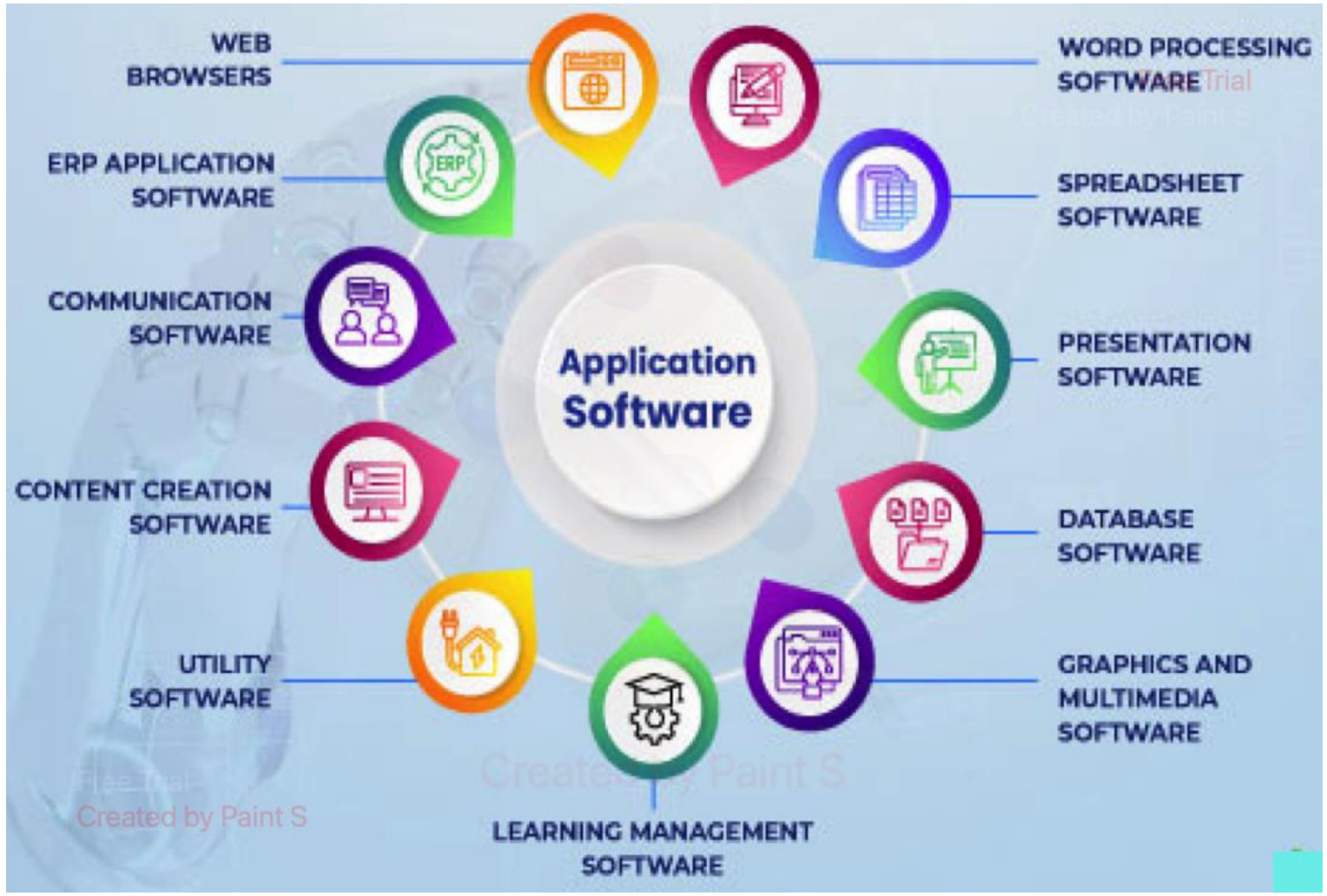




Application Software

- also known as end-user programs or productivity programs
- performs special functions or provides functions that are much more than the basic operation of the computer
- designed to perform a specific task for end-users
- help the user in completing tasks such as doing online research, jotting down notes, setting the alarm, designing graphics, keeping an account log, doing calculations, or even playing games
- can be referred to as non-essential software as their requirement is highly subjective, and their absence does not affect the system's functioning

Types of Application Software



Application Software Types

Word Processing Software: Word Processing Software can be explained as software that has the functionalities of editing, saving, and creating documents with Word Processor Software like Microsoft Word.

Presentation Software: It is a type of application software that is used to present some applications like newly launched functions, products, etc. For Example, Microsoft Powerpoint.

Educational Software: Due to the enhancement of the Internet, there are so many educational software runs in the market. It consists of Language learning Software, Classroom Management Software, etc.

Database Software: A database management system is an application software to create and manage databases. They help with the organization of data

Spreadsheet Software: Spreadsheet Software is a kind of software that deals with the worksheet where it works on some automated version to perform numeric functions. For Example, Microsoft Excel.

Multimedia Software: Multimedia refers to the mixture of audio, video, image, text, etc., and can be displayed or used with the help of multimedia software. There are so many media players that do this kind of work.

Web Browsers: Web Browser is one of the most used applications worldwide, it takes you to the internet. You can use your desktop, mobile, etc for using this.

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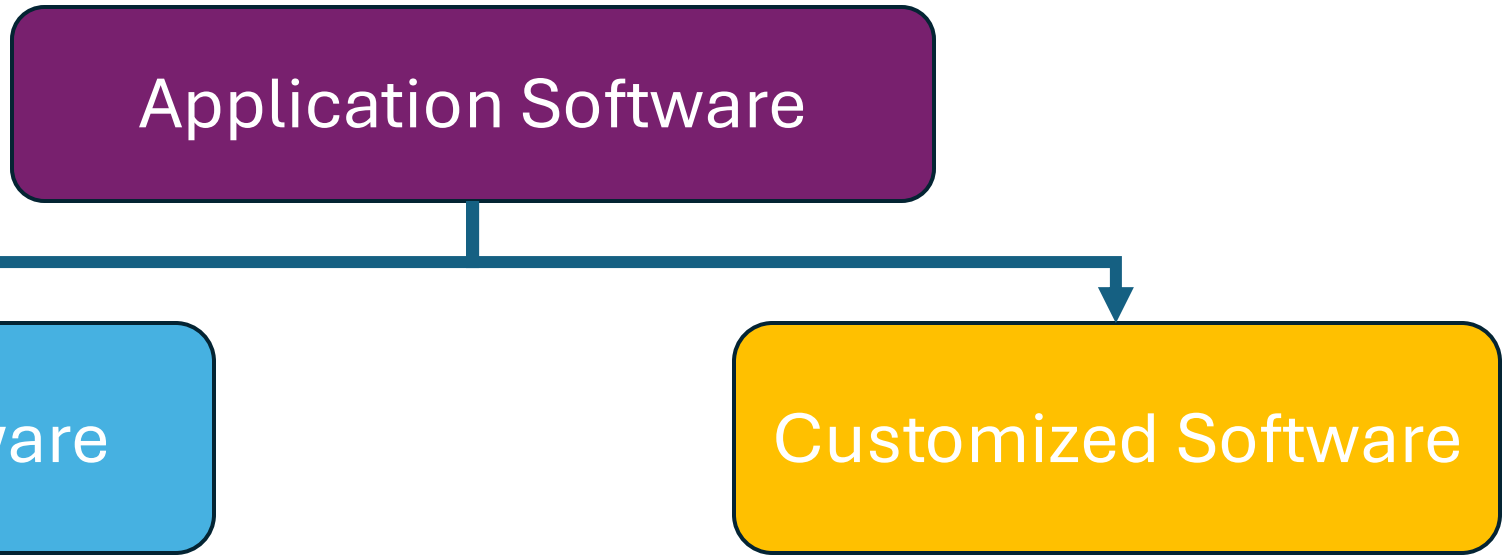
Application Software Types



Programs prewritten by professional programmers that are typically offered for sale.

Programs written for a specific purpose and for a specific organization.

Generic Software (packaged software)

- A system designed for general public usage.
- Isn’t created for a specific purpose, organization, or individual.
- functionality designed by software developers to solve a particular problem that will be used by many entities.
- created for all types of business requirements which has a positive demand in the open market.
- The developers create this type of software by themselves and handle it for a group of clients who have similar project requirements.

Generic Software (packaged software)

Stand-alone systems that are produced by a development organization and sold on the open market to any customer who is able to buy them.

- Examples:
  - Office Packages (Microsoft Office, Open Office etc)
  - Drawing Packages (AutoCAD, Adobe Photoshop …)
  - Databases (MySql, MSSQL)
  - …

Customized Software (bespoke software)

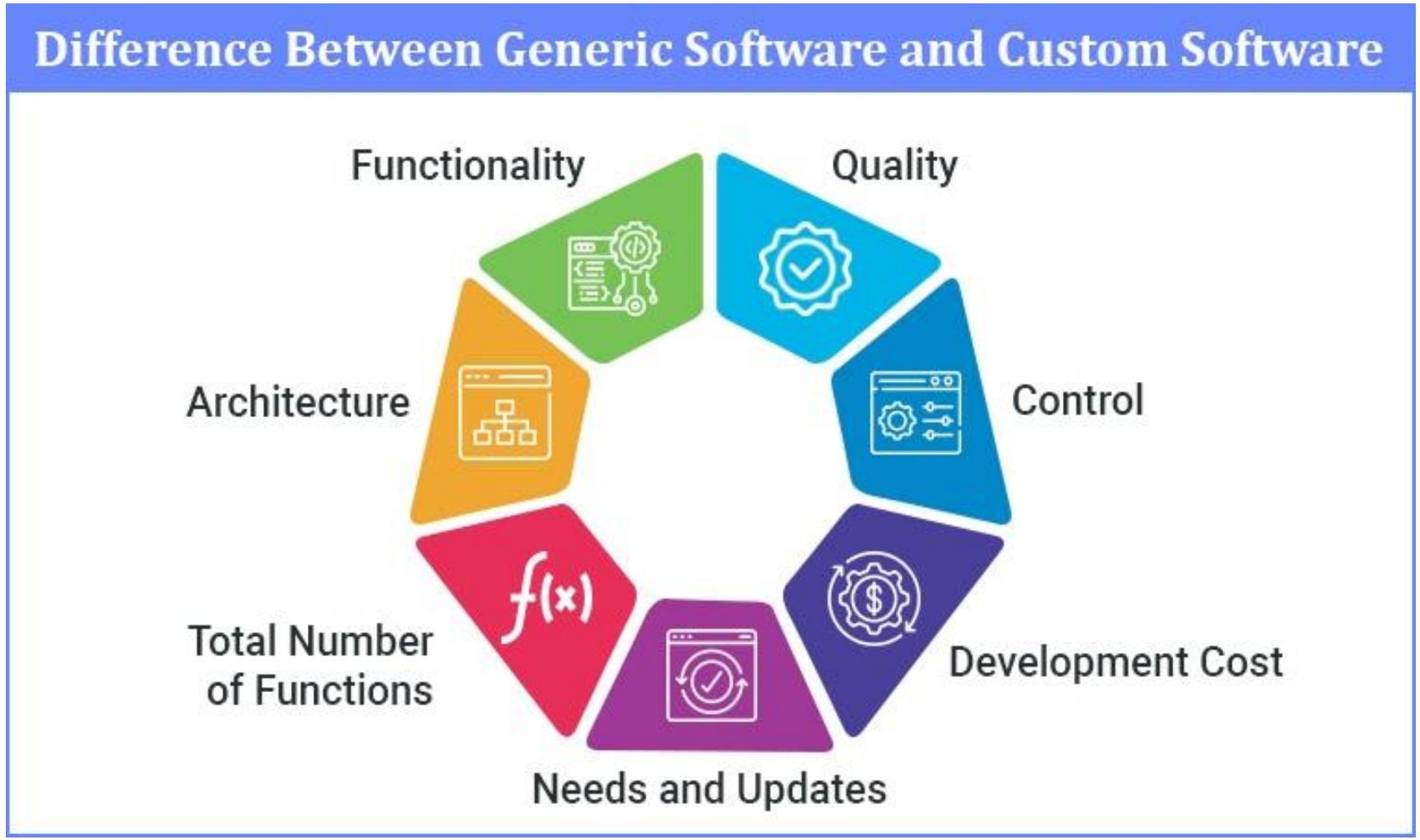
- a specifically designed system for a particular customer or an organization.
- a business organization or a user hires a software development company to create software that can offer some specific results and functionalities.
- not available for general use only the customer for whom it is available has the right to use its distinctive features.
- It only caters to the requirements of a specific firm which allows it to gain a competitive advantage over the market.

Customized Software (bespoke software)

Developed for a single customer according to their specification.

- Examples:
  - Control systems for electronic devices
  - Air traffic Control systems
  - Healthcare systems
  - Payroll systems
  - …

Generic Vs. Customized Software



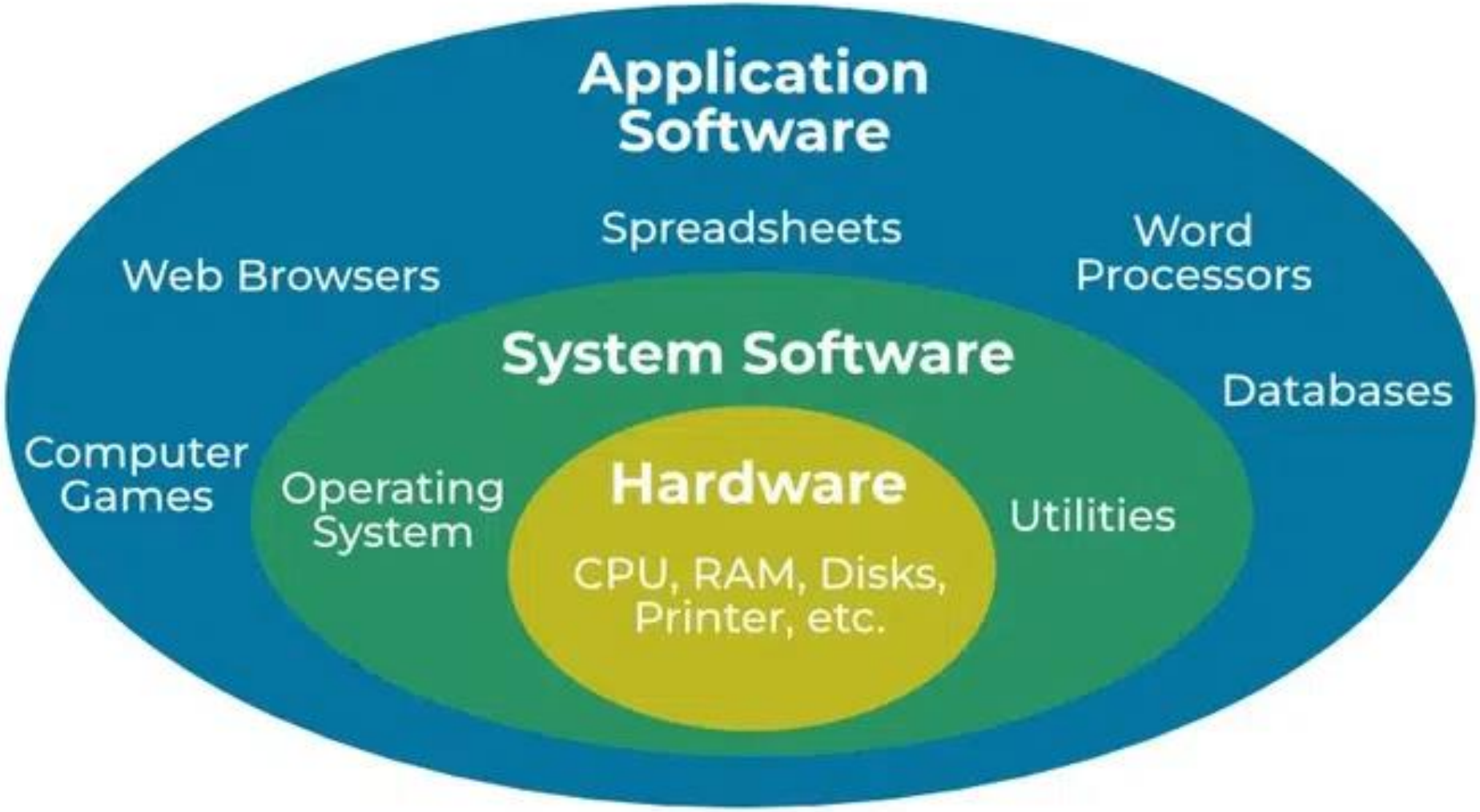
Generic Vs. Customized Software

	Generic Software	Customized Software
Quality	Less concern	High concern
Control	Developers & Designers has the power	Customer has the power
Development Cost	Comparably low cost	Comparably high cost
Needs and Updates	As per the market requirement	As per the customer requirement
Total number of Functions	Wide number of functions	Functionas related to specific entity
Architecture	Scalable, user friendly and agile	Balance between client requirements and scalability
Functionality	Multiple functions that are common	Functionality as per the business requirements

Activity

- Identify customized products and generic products from the following list
  - Adobe Photoshop
  - Microsoft SQL Server
  - Rent A Car system
  - VLC Player
  - Airline Reservation System
  - Pharmacy Management System
  - Python
  - Library Management System

System Vs. Application Software



System Vs. Application Software

System Software	Application Software
It is designed to manage the resources of the computer system, like memory and process management, etc.	It is designed to fulfil the requirements of the user for performing specific tasks.
Written in a low-level language.	Written in a high-level language.
Less interactive for the users.	More interactive for the users.
System software plays vital role for the effective functioning of a system.	Application software is not so important for the functioning of the system, as it is task specific.
It is independent of the application software to run.	It needs system software to run.

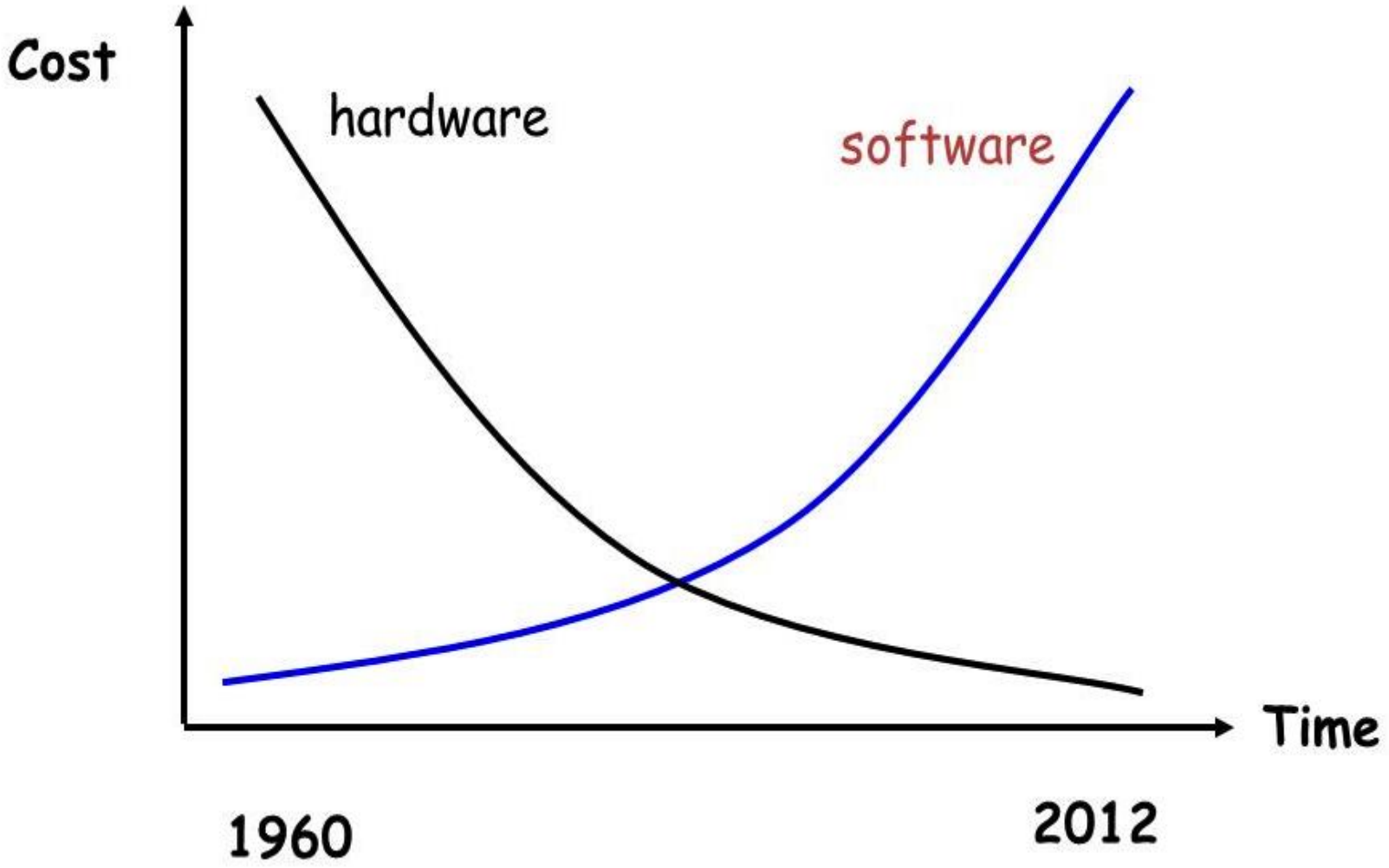
Software Projects vs. Other Projects

Software Projects vs. Other Projects

- The main difference in software engineering compared to other engineering disciplines are as follows;
  - It is difficult for a customer to specify requirements completely.
  - It is difficult for the developer to understand fully the customer needs.
  - Software requirements change regularly.
  - The environments to which software is made change regularly.
  - Software is primarily intangible; much of the process of creating software is also intangible, involving experience, thought and imagination (a brain product).
  - Because of intangibility it is difficult to specify software.
  - Customers and developers have communication gaps.
  - It is difficult to test software exhaustively.
  - Not like other engineering disciplines that have narrow scope, software can be developed in diverse contexts ranging from embedded systems to giant manufacturing systems.
  - A small change to a few lines of code could do a big change in the overall behaviour of the system.

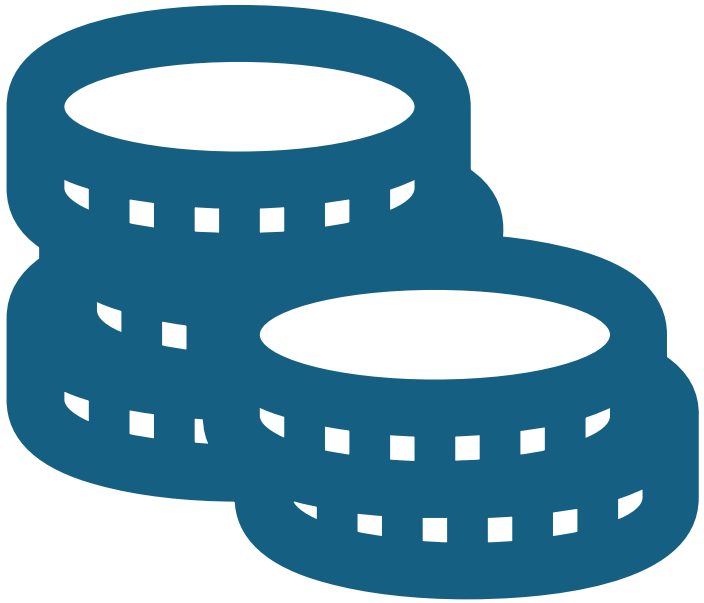
Software & Hardware costs

Cost of Harware & Software



Software Costs

- Software costs often dominate computer system costs.
- The costs of software are often greater than the hardware cost.
- Software costs more to maintain than it does to develop.
- For systems with a long life, maintenance costs may be several times of development costs.
- Software engineering is concerned with cost effective software development



Activity

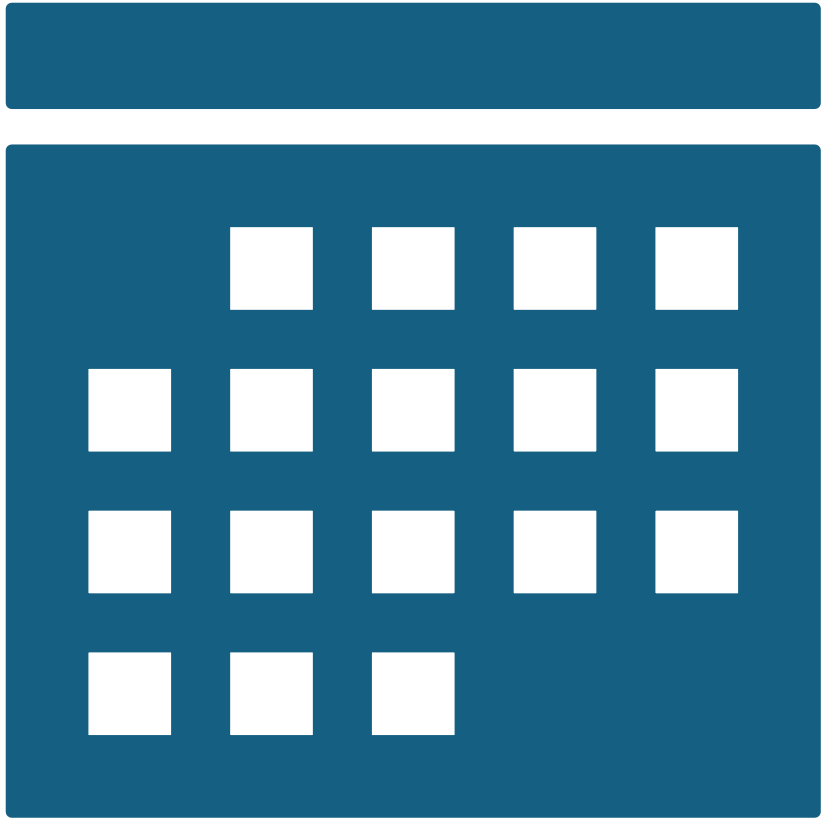
- Identify and list down factors affecting software development costs

Software cost Factors

- Scope of the Project
- Complexity of the Project
- Size and Location of the App Development Team
- Experience and Expertise of the Development Team
- Use of Open-Source Technologies or Third-Party Tools
- Timeline of the Project
- Quality of the Project
- Ongoing Maintenance and Support

Software Cost Estimation

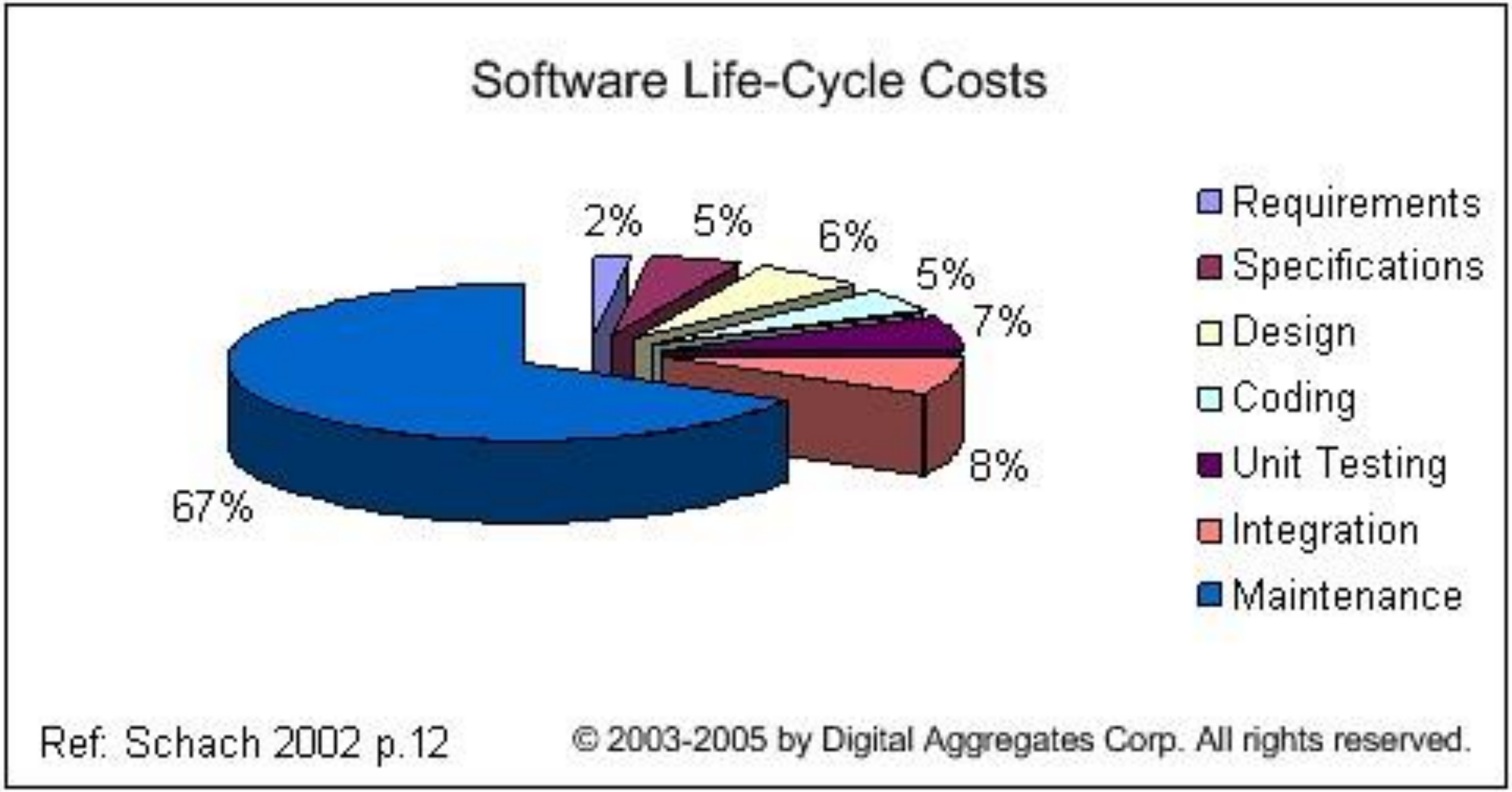
- How much effort is required to complete each activity?
- How much calendar time is needed to complete each activity?
- What is the total cost of each activity?



Software Development Cost

The total cost of a software development project is the sum of following costs:

- Hardware and software costs including maintenance.
- Travel and training costs.
- Effort costs of paying software developers.


Software Life-Cycle Costs
2%
5%
6%
5%
7%
8%
67%
Requirements
Specifications
Design
Coding
Unit Testing
Integration
Maintenance
Ref: Schach 2002 p.12
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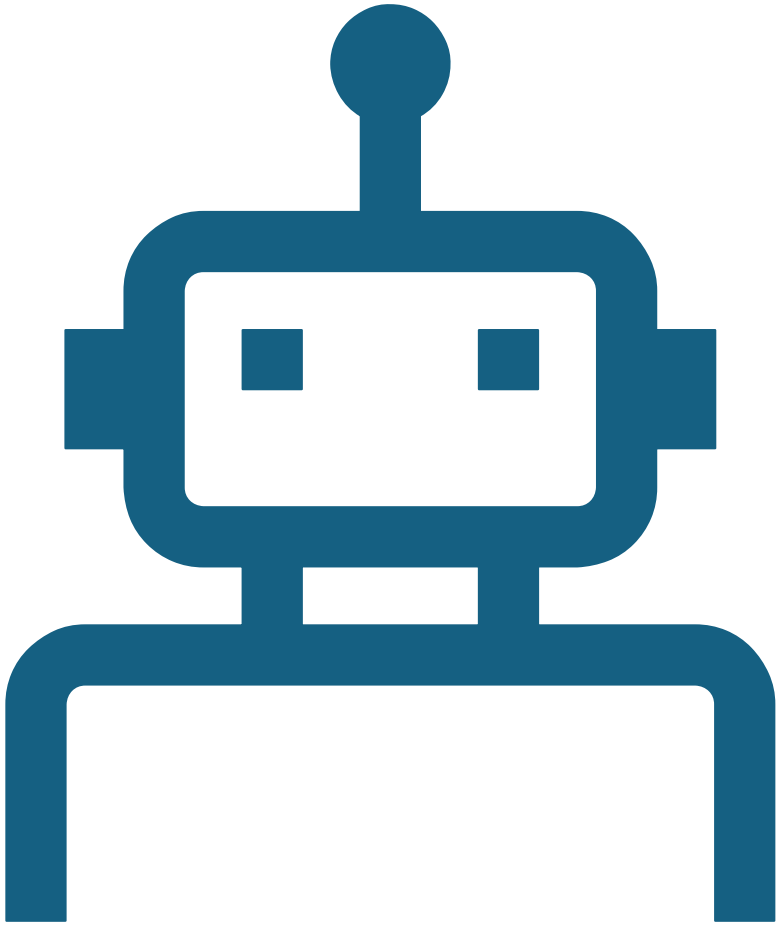
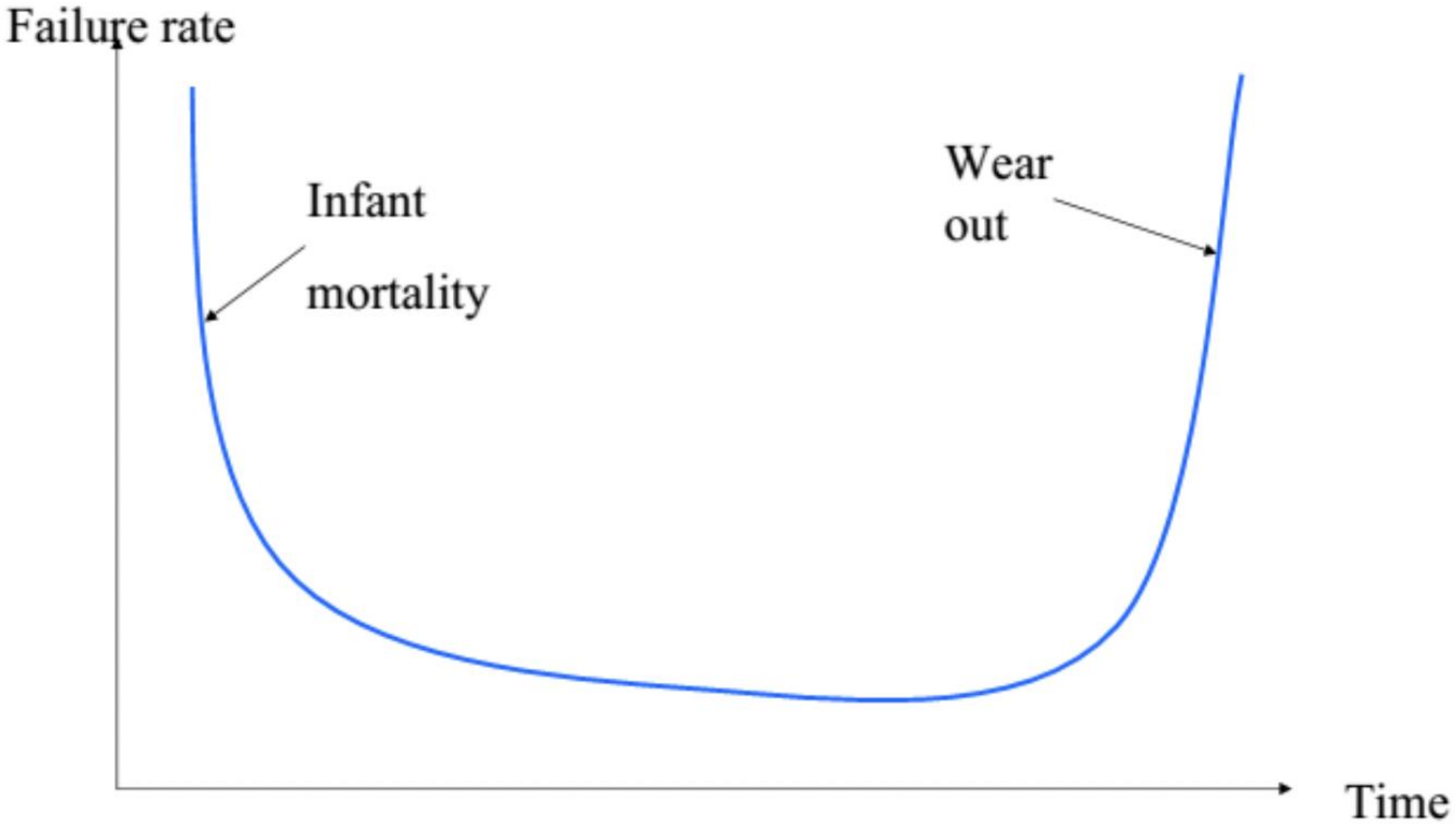
Software & Hardware Failures

Failure curves for Software & Hardware

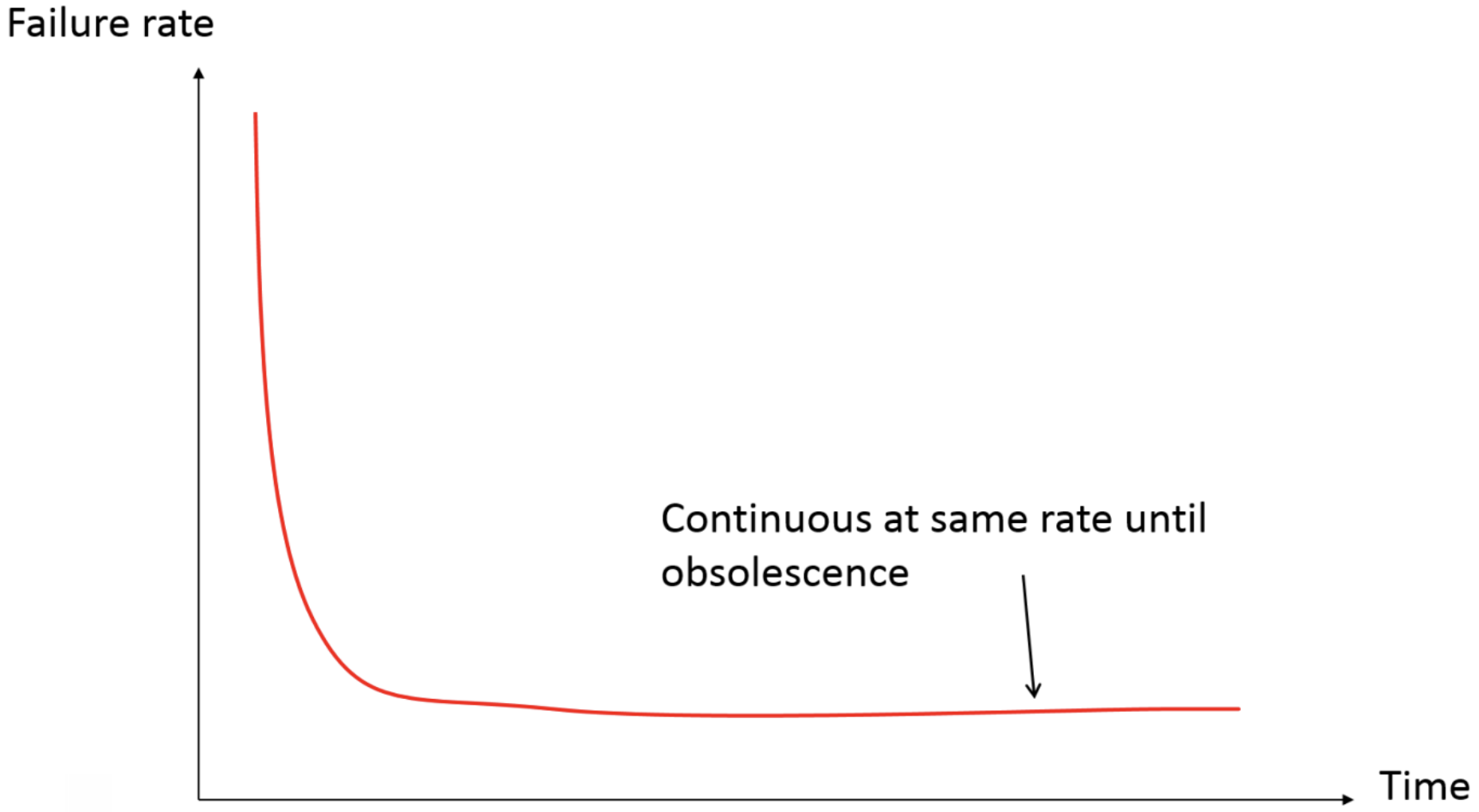
- Software is engineered or developed, not manufactured in the traditional sense.
- Software does not wear out in the same sense as hardware.



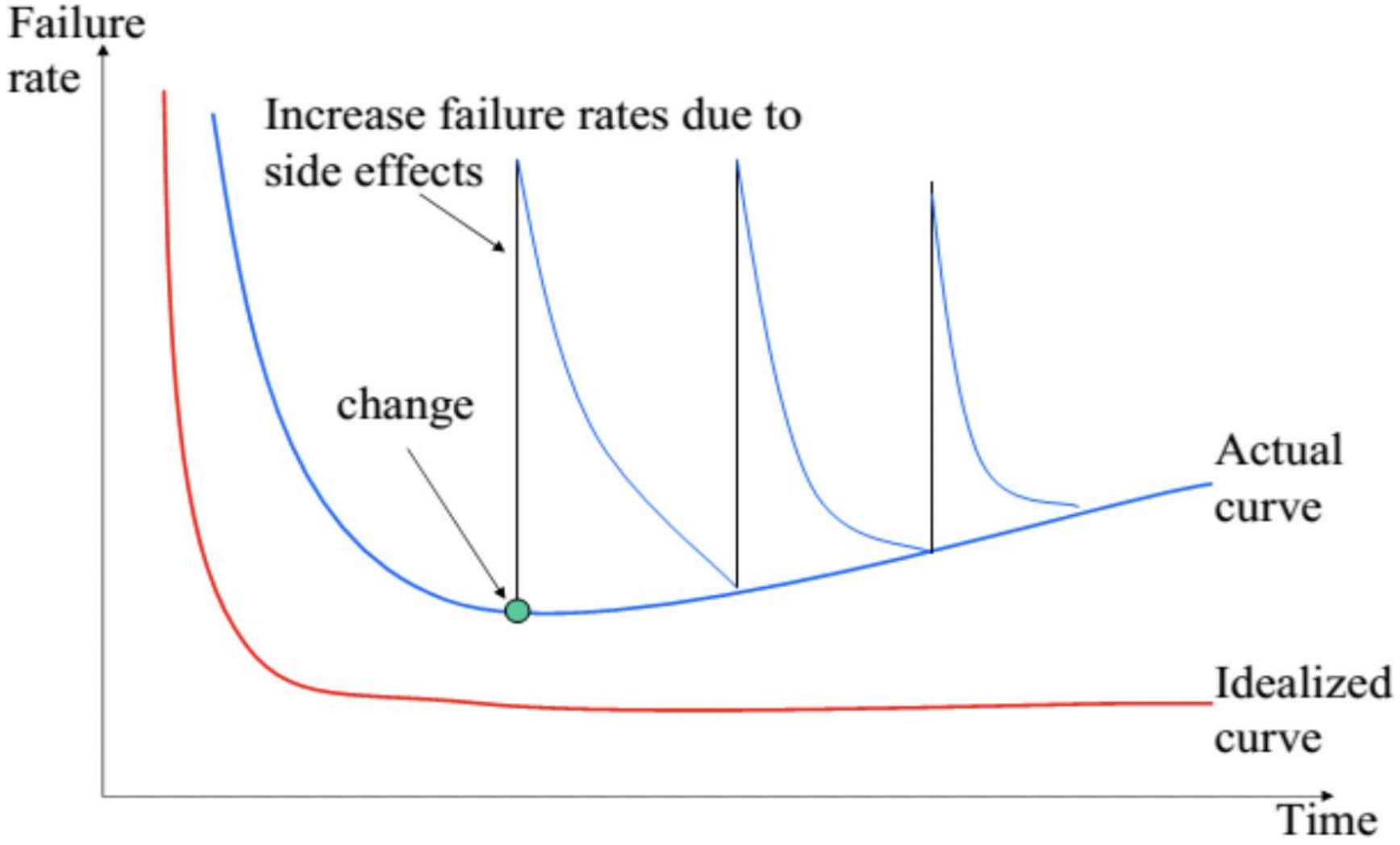
Failure ‘Bathtub’ curve for Hardware



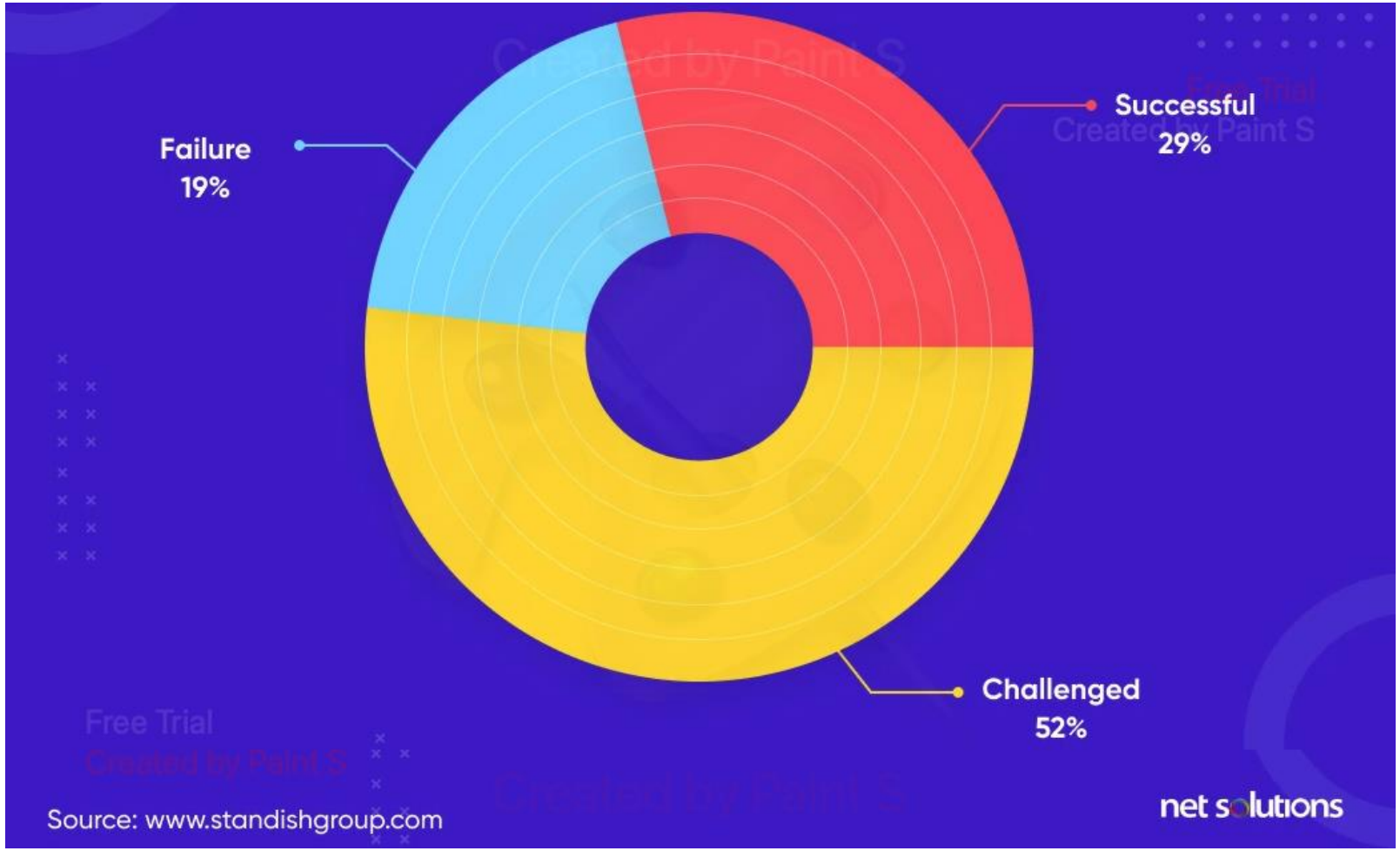
Failure curve for Software



Failure curve for Software: in reality



Success rate of Software Projects

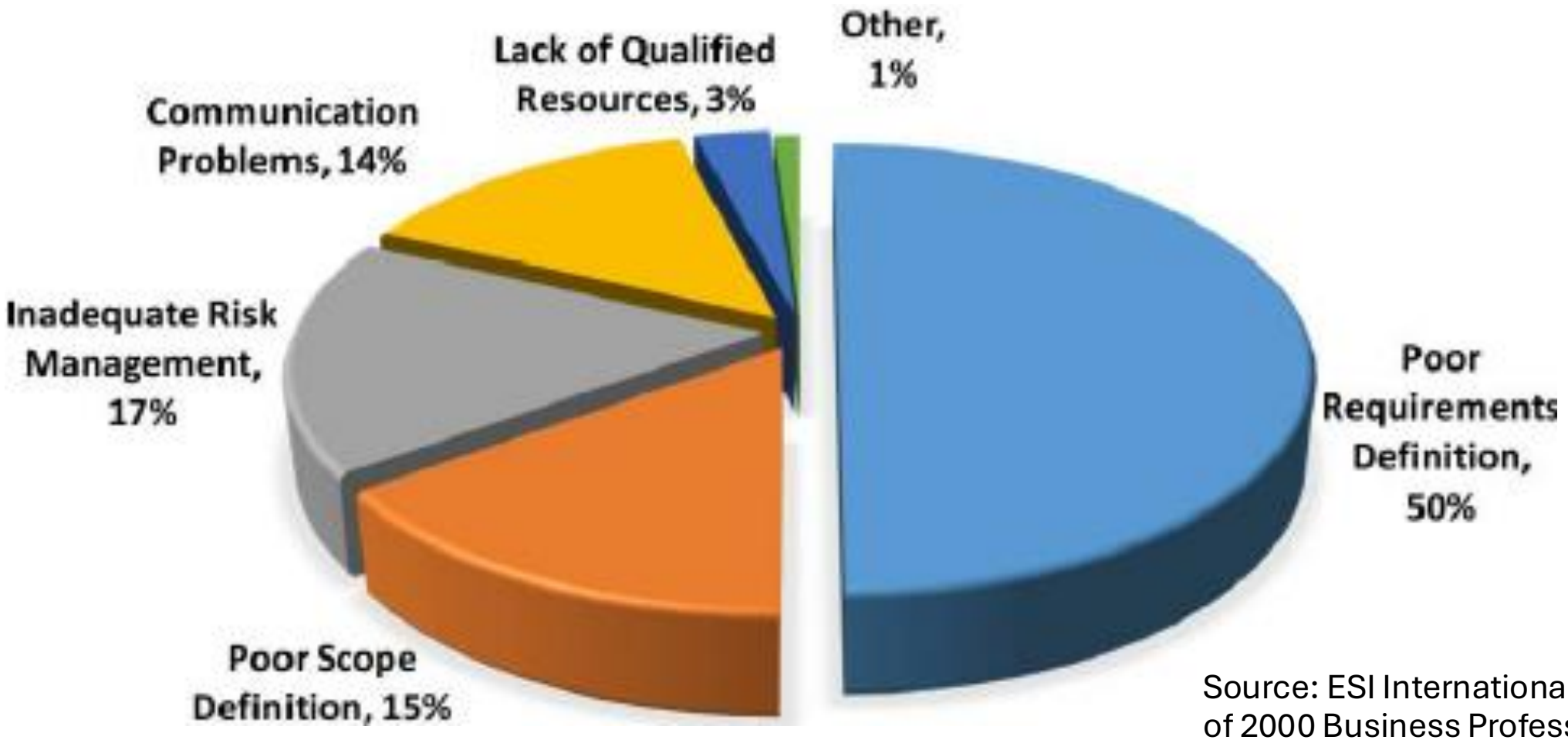


- Successful – ontime, on budget, with all the specified features and functions
- Failure – cancelled before they were completed
- Challenged – Deliverd and operational, but over budget, over schedule or with fewer features

Activity

- Identify and list down reasons for software failures

Reasons for Software Failure



Source: ESI International Survey of 2000 Business Professionals, 2005.

Software Disasters

Software Disasters

"To err is human, but to really foul things up you need a computer"
- Paul Ehrlich

- Over the years, there have been several major software disasters, resulting from poor software project management, poor risk assessment, and poor development and testing practices.

- The results of the disasters range from project delays, project cancelations, loss of millions of dollars of equipment, to human fatalities.

- It is important to study software disasters, to alert developers and testers to be ever vigilant, and to understand that huge catastrophes can arise from what seem like small problems.

Activity

- SEARCH FOR SOFTWARE DISASTERS
- SELECT ONE AND PRESENT TO THE CLASS